

EXPERIMENT 1: IMPLEMENTATION OF CAESAR CIPHER

AIM:

To implement the simple substitution technique named **Caesar Cipher** using C language.

ALGORITHM:

1. Start the program.
2. Read the plaintext message from the user.
3. Read the key value (shift value).
4. For each character in the plaintext:
 - If it is an uppercase letter, shift it by the key value.
 - If it is a lowercase letter, shift it by the key value.
 - Leave spaces and special characters unchanged.
5. Display the encrypted ciphertext.
6. Stop the program.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char text[100];
```

```
    int key, i;
```

```
    printf("Enter the plaintext: ");
```

```
fgets(text, sizeof(text), stdin);
```

```
printf("Enter the key: ");
```

```
scanf("%d", &key);
```

```
for (i = 0; text[i] != '\0'; i++)
```

```
{
```

```
    if (text[i] >= 'A' && text[i] <= 'Z')
```

```
        text[i] = (text[i] - 'A' + key) % 26 + 'A';
```

```
    else if (text[i] >= 'a' && text[i] <= 'z')
```

```
        text[i] = (text[i] - 'a' + key) % 26 + 'a';
```

```
}
```

```
printf("Encrypted text: %s", text);
```

```
return 0;
```

```
}
```

OUTPUT:

Output Clear

Enter the plaintext: Kanmani
Enter the key: 6
Encrypted text: Qgtsgto

=== Code Execution Successful ===

RESULT:

Thus the implementation of Caesar cipher had been executed successfully.